

Refractor: Steerable Live Clustering for Speaker-Audience Understanding

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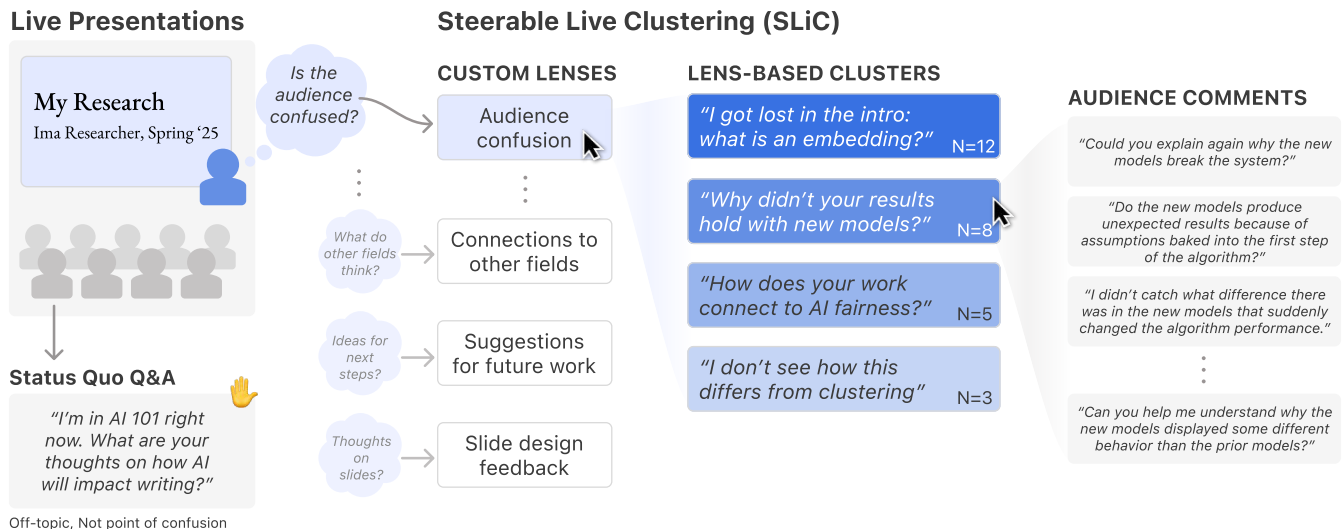


Figure 1: (Left) Live talks can lead to interactions that are missed opportunities for speakers to learn from their audiences. (Right) We introduce *steerable live clustering*, an incremental LLM-based clustering method that organizes clusters of audience comments with respect to flexible speaker-specified text inputs (*lenses*). Each cluster is represented in the interface by a quote from the audience.

Abstract

Current audience-speaker interaction methods flatten audience feedback onto one lens—i.e., via the first raised hands, or the most-upvoted comments—but what if an educator instead wants to focus on areas of student confusion, or a researcher wants to seek critiques from external fields? To support this plurality of perspectives, we introduce an audience interaction tool called Refractor that allows speakers to define custom lenses to understand their audience. For example, a speaker can specify an “urgent questions” lens during a live presentation, and the system will display a high-level synthesis of time-critical misunderstandings from the audience. After the talk, the speaker can apply an “interdisciplinary critiques” or “suggestions to expand work” lens to explore the audience’s perspectives on future work. To instantiate Refractor, we introduce

Steerable Live Clustering (SLiC), an algorithm that uses speaker-specified lenses to rapidly prioritize and thematically group text comments at interactive speeds.

CCS Concepts

• **Human-centered computing** → *Collaborative and social computing systems and tools; Interactive systems and tools; Visualization systems and tools.*

Keywords

live presentations, audience response systems, unstructured text analysis, topic modeling, large language models, human-AI interaction

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1 INTRODUCTION

Many pivotal acts of communication happen via live presentations; the success of a presentation critically depends on an understanding of the audience at hand. However, there is not just one way to understand an audience: while an educator might prioritize student comments expressing confusion, a PhD student delivering a job talk may want to be better equipped to address critiques from other subareas. However, audience interaction methods today prescribe one dominant lens into perceiving the audience that may or may not be aligned with a speaker’s interests. From polling tools like Slido and PolleEV [9, 11] that only foreground popular comments to traditional hand-raising which offers narrow insight into select audience members, the status quo of audience interaction flattens the many possible ways of interacting with audience opinions to singular “default” options. What if our audience interaction tools could move beyond a single lens and instead support a plurality of viewpoints from which a speaker may wish to understand their audience?

Previously, it was not possible to dynamically analyze audience comments via thematic, high-level goals. Traditional text analysis methods such as topic models [4, 10] and live tag clouds [12] run quickly, but operate at the keyword level (e.g., “health,” “questions”) and cannot capture high-level audience ideas (e.g., “critique of healthcare framing,” “student confusion about notation”). Further, several prior works show that text analysis tools require substantial adaptation to serve domain-specific applications from education [3, 5, 6] to social media [15, 16] to online recipes [1]. Advances in large language models (LLMs)—with broad world knowledge and strong language comprehension—have unlocked more flexible unstructured text analysis tools that can capture these diverse, high-level priorities [7, 8, 14]. However, while LLM-based clustering methods are broadly steerable, they operate slowly—on the order of minutes rather than seconds—due to the multiple iterations required to make them reliable.

To address this challenge, we introduce a novel algorithm we term *Steerable Live Clustering* (SLiC) that centers user-specified goals, which we call *lenses*, to thematically group text comments at interactive speeds to dynamically reflect input comment streams. Drawing on prior work [2, 8], we achieve this speed by implementing SLiC as a streaming algorithm that can run continuously via per-comment operators rather than requiring a complete dataset. To enable steerability, SLiC allows speakers to input a lens that shapes the algorithm’s clusters: for example, a PhD student delivering a practice job talk could specify a “high-level narrative” lens to get a sense of how the audience resonates with the overarching vision, and then a “slide design” lens to dig into more detailed slide-level feedback.

We demonstrate our vision of steerable audience understanding by incorporating the SLiC algorithm into Refractor, an interactive system for organizing audience comments. Our work presents a novel approach to audience interaction that fundamentally centers speaker goals—from low-level algorithm operators to high-level interface controls.

2 SYSTEM

The Refractor system allows speakers to flexibly explore audience feedback through custom lenses. The system consists of (1) the Steerable Live Clustering algorithm (SLiC) that instantiates speaker-authored lenses to prioritize and group audience comments and (2) a web interface that applies SLiC to synthesize and visualize audience feedback.

We introduce the Steerable Live Clustering algorithm (SLiC) to address the unique information synthesis needs of live presentations. In particular, we focus on *steerability*, the need to align with speakers’ specific analytic lenses, and *speed*, the need to surface clusters rapidly in real-time. While LLM clustering approaches are steerable, LLM-based clustering algorithms mitigate base prompting reliability issues by requiring multiple passes [7, 8, 14], leading to slower outputs unsuited for live talks. Thus, building on these recent approaches, our clustering algorithm focuses on two key design decisions that preserve reliable open-ended steering, but at interactive speeds: our clustering operator, which groups comments on a per-comment basis, and our score operator, which scores comments for their alignment to the user’s inputted speaker preference, which we call a *lens*. Some examples of prioritized clusters produced by different lenses can be seen in Figure 2.

2.0.1 Clustering Comments. Our first step is to produce cohesive clusters to capture audience perspectives subject to the given lens and presentation context via the `gen_cluster` operator. As input to this operator, we include the speaker lens l and a presentation document. This speaker-provided context helps the LLM to (1) better interpret audience comments and (2) create clusters relevant to the speaker’s desired lens.

This clustering method builds on the TopicGPT’s topic generation algorithm, which iterates over a sample of documents from the original corpus to generate cluster labels [8]. We iterate through the corpus of audience comments C and apply the `gen_cluster` operator to each comment c . This operator prompts an LLM (GPT-4o) to either (1) assign c to an existing cluster in the set of clusters S , or (2) create a new cluster that better describes c and add it to S . Notably, while TopicGPT performs topic generation on a sample of the dataset before topic assignment on the full dataset, our algorithm supports continual topic generation. This change is critical because in the live setting, the system only has access to an incomplete set of comments. Different topics and trends continuously emerge as the talk unfolds, so it is critical that new comments streaming into the algorithm can inform cluster creation.

2.0.2 Prioritizing Lens-Aligned Comments. Although specifying a lens is helpful, it is often the case that many comments will not be related to the lens the speaker seeks. After clusters assignment, we prioritize the comments most aligned with a speaker-specified *lens* (e.g., “feedback on visual design,” “feedback on structure of the presentation”) by scoring each comment’s alignment to the provided lens l . The score operator produces this lens-relevance score by prompting an LLM (GPT-4o) to evaluate each comment on a scale from 0 (“does not contain speaker’s requested feedback”) to 2 (“demonstrates speaker’s requested feedback clearly”). This function operates independently of the `gen_cluster` function, and determines the relative weighting for each cluster. Clusters most

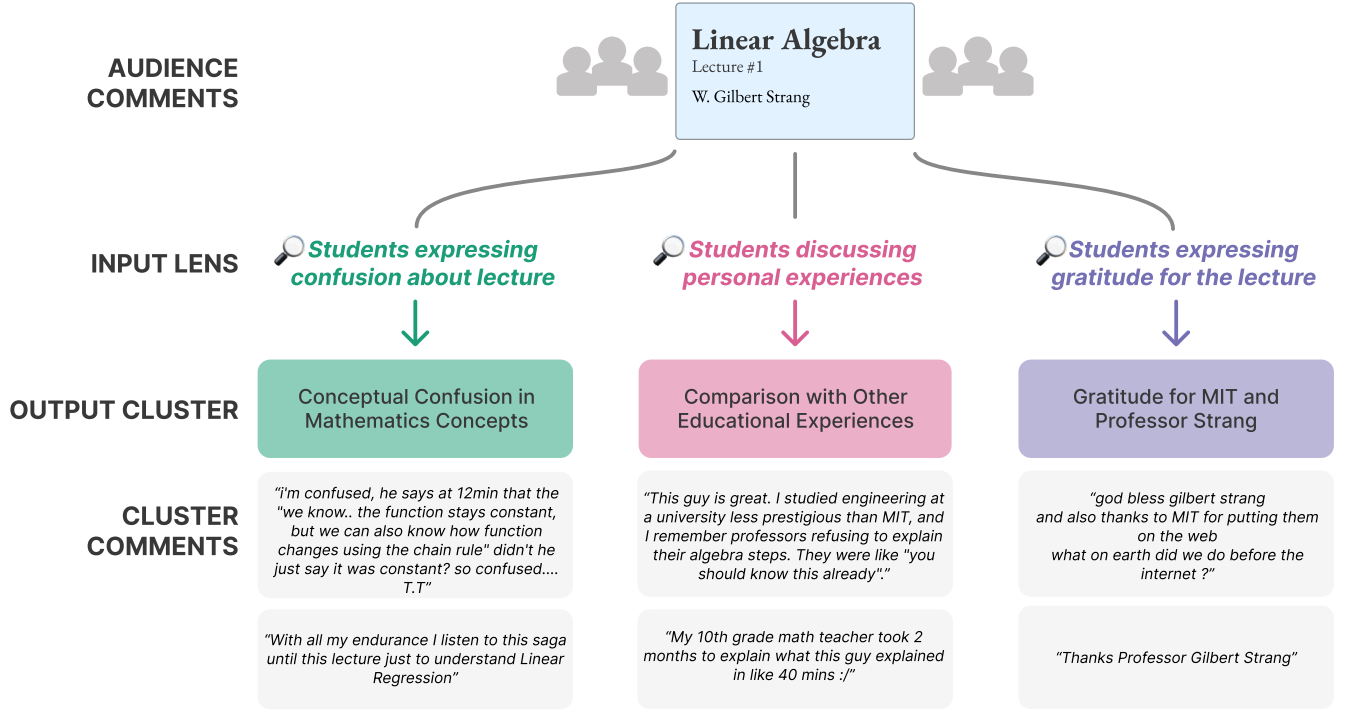


Figure 2: We display examples of the prioritized clusters and comments that result from using different seeds. The comments displayed are from the SIGHT dataset [13], a collection of Youtube comments left on MIT OpenCourseWare videos.

aligned with the speaker lens are subsequently highlighted in the interface.

2.0.3 Visualizing Clusters for Speakers. We surface these clusters (e.g., Figure 2) in a visual interface (Figure 3) that aids speakers in quickly skimming and prioritizing clusters to explore comments. We walk through the Refractor comment-viewing interfaces in greater detail below.

Cluster Aggregate View. As comments are submitted, the Cluster Aggregate View (Figure 3B) presents a simplified, high-level overview of audience comments. Each cluster is represented as a color-coded box that displays one representative quote for the cluster. We prioritize clusters with a composite score that accounts for both the speaker lens and audience prevalence. The speaker score S_s is the average lens-relevance score for comments in a given cluster (generated by the score operator). The audience score S_a captures how popular the cluster topic is among audience members. Then, the cluster prioritization score is defined as: $W_a \cdot S_a + W_s \cdot S_s$, where W_a , W_s are importance weights assigned to the audience score S_a and the speaker score S_s , respectively. Users can interactively adjust these weights with controls in the Refractor interface. This cluster prioritization score determines the color value of each cluster in the Refractor interface so that higher-priority clusters are darker and more visually salient. Beyond focusing attention on a smaller set of clusters, we simplify each cluster to an overview to further aid skimmability. We use the `select_quote` operator (which prompts GPT-4o-mini) to select a single quote to represent

the entire cluster, which we find provides a succinct but grounded sense of the audience perspectives in that cluster.

The control panel (Figure 3A) allows users to customize the Cluster Aggregate View to their own preferences. For example, users can modify the cluster sorting method, or they can alter visual cluster highlighting by adjusting the relative weights of speaker alignment scores and audience popularity.

Cluster Detail View. While the Cluster Aggregate View presents an overview of all audience comments, the Cluster Detail View displays all of the comments for an individual cluster upon user selection (Figure 3C). This view allows users to explore a particular cluster of interest in depth.

2.0.4 Other System Details. Refractor includes Speaker and Audience Onboarding Views where users can login to the interface and receive or add comments respectively for a given talk.

The system combines a SvelteKit frontend with a Flask backend, using Firebase to store comments, speaker data, and clustering results, and Redis to manage active jobs. The live version uses incremental steerable live clustering: the frontend polls the backend for real-time updates until comments close, with results stored as a JSON.

3 CASE STUDIES ON LIVE DEPLOYMENT

Next, we investigate how Refractor performs on real audience comments in a series of live deployments (Table 1). These deployments provide evidence that Refractor can be useful for guiding post-talk

SPEAKER SYSTEM OVERVIEW

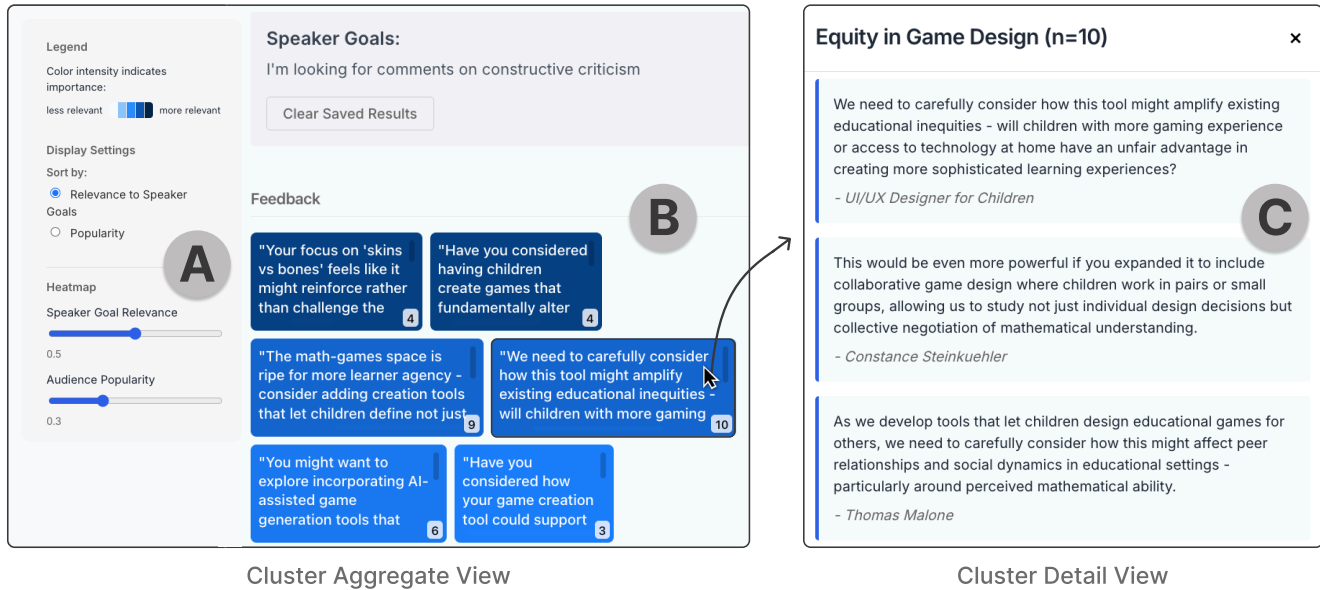


Figure 3: The Refractor web app helps speakers collect and flexibly cluster feedback from audience members. (Left) The Cluster Aggregate View displays a high-level overview of audience comments by representing each cluster with a different box. The legend (A) displays user settings for personalizing the interface, whereas the clusters themselves are on the right side of the page (B). Clicking on one cluster reveals the Cluster Detail View (C), which pops up when a user clicks into a cluster.

discussions from an audience as large as 130 students to as small as 4 researchers.

4 CONCLUSION

While current audience interaction methods focus on one lens into perceiving feedback, we argue that these systems should support the diverse perspectives and interests that speakers have. We introduce Refractor, an audience interaction tool that supports speaker-specified lenses for understanding audience comments, offering speakers a new way to exert their preferences over how they make sense of feedback.

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Setting	Deployment Frequency	Speaker Goals	High-level Results
PhD student presenting their research at an in-person lab meeting of four people.	One-time	Finding feedback on research ideas.	Refractor helped the speaker triage comments more relevant to their goals, even with a smaller audience size.
University student presenting their startup pitch to an audience of six students.	One-time	Finding feedback on communication style and clarity of pitch.	Audience members appreciated the tool as a scaffold for post-talk discussion and the speaker found that this tool led to greater transparency during discussions.
Instructor teaching a class on HCI Research with 130 students. The tool was used on student discussion responses to assigned readings.	Twice a week for ten weeks	Highlighting unique student ideas that do not echo arguments already found in the readings.	Refractor allowed the instructor to perform live synthesis of student commentaries during lecture time, even when the responses were due less than an hour before the class started.

Table 1: Overall, these case studies provide evidence that Refractor can be useful for reflecting on and responding to audiences as large as 130 students to as small as 4 researchers.

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